1 The problem

All four reactor experiments in CHE331 (Batch, CSTR, PFR and RTD) are run at fixed conditions: one reactor, three flow rates, three temperatures and equal dilute feeds of NaOH and ethyl acetate. Readings are written down and analysed later, so nobody sees what is happening inside the reactor while the rig is running. Questions such as why conversion falls with flow, what a dead zone does, or why one residence time is not enough to reach steady state are answered in words, never tested. In our own CSTR run the conductivity was still drifting when the "steady" readings were taken, and 28 °C gave more conversion than 40 °C. We had no way to find out why.

The proposal set out to build one web app, used on a laptop next to the rig with no new hardware, that turns the manual readings into proper fits, live plots, assumption checks and what-if studies.

2 What I built

The live app is at `https://che331.streamlit.app`.

2.1 Testing

The models and pages have been tested by me on a range of analytic checks and on the CHE331 lab-report readings. The full test suite is on GitHub: ReactorGenome / tests.

2.2 CSTR

Assumptions (equimolar feed, ideal mixing, isothermal, irreversible) sit as switches on the left. Readings are typed exactly as on the lab sheet: one row per run, plus a small Λ_0 table per temperature (Figure 2).

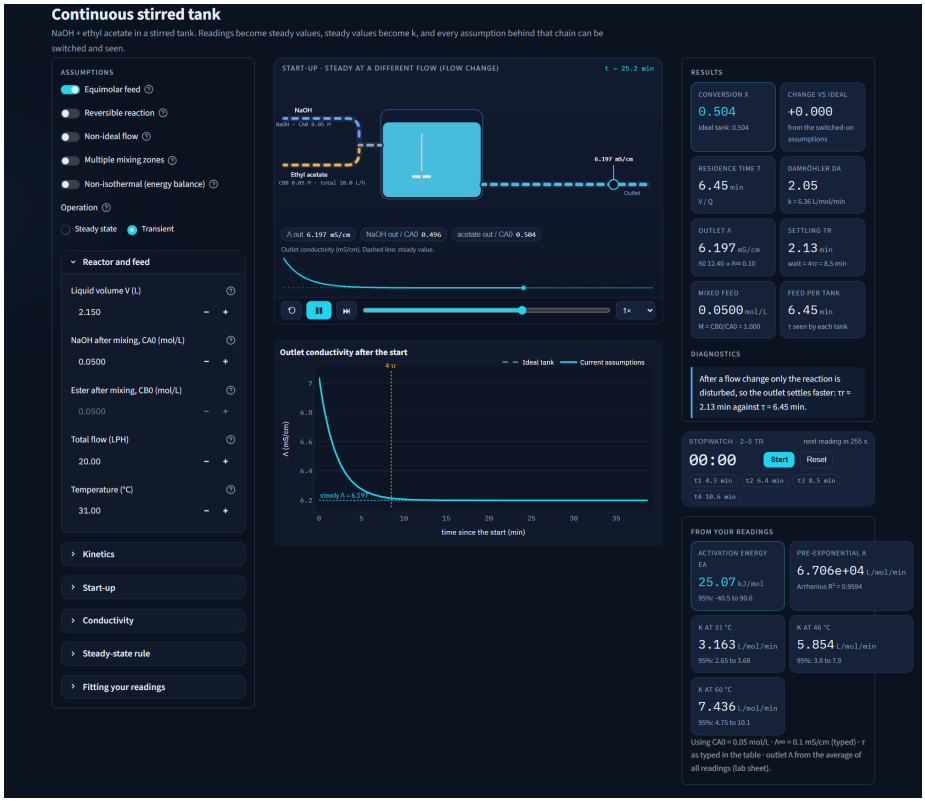


Figure 1: CSTR page: assumption switches, animated start-up, results, and fitted kinetics ($E_a = 25.07\,\mathrm{kJ/mol}$).

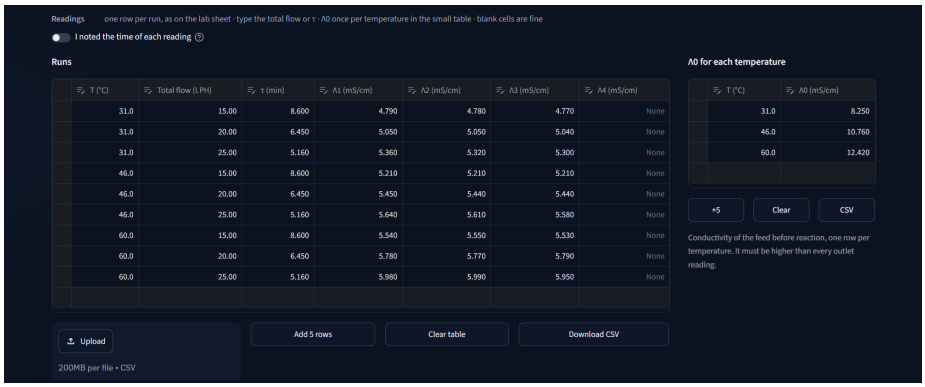


Figure 2: Readings table, laid out like the lab sheet.

The Results tab puts everything the lab report needs on one screen: E_a, A, k at each temperature with 95% intervals, design plots, Arrhenius, conversion against flow and temperature, single-run k against τ, parity and residuals (Figure 3). The rate-law order (1 to 3) is a setting.



Figure 3: Results tab: kinetics and all of the report plots on one screen.

The fitted $k(T)$ then predicts X at any temperature, feed concentration and τ. The right-hand plot shows why one concentration cannot pin down the order: every order fits the runs at the C_{A0} used, and the orders separate only away from it (Figure 4).

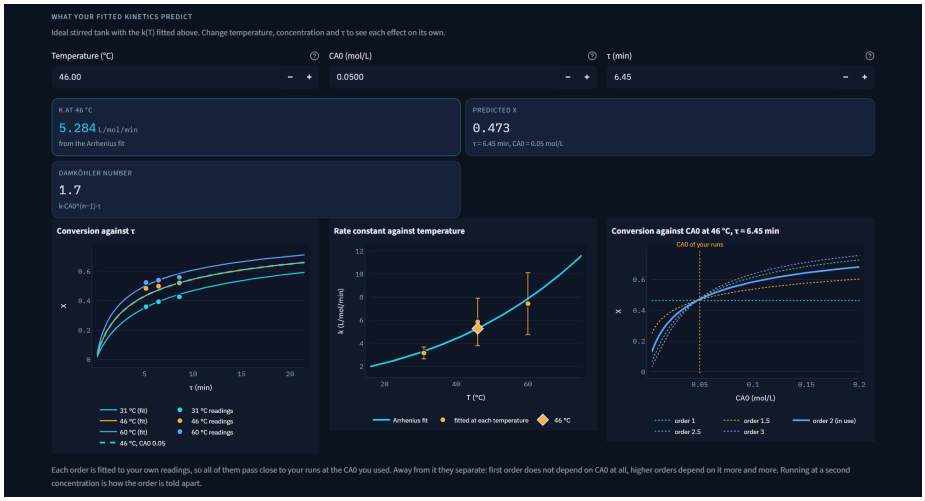


Figure 4: Predictions: X against τ, k against T, and X against C_{A0} for orders 1 to 3.

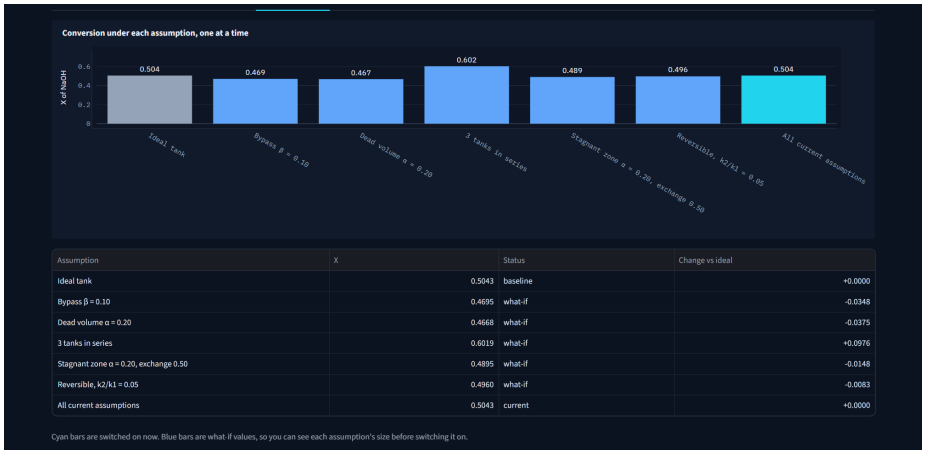


Figure 5: Assumption effects: conversion with each textbook assumption dropped on its own.

The energy-balance tab is the scale-up case from the proposal. With a concentrated feed and a weak jacket (Figure 6) the heat-generation curve crosses the heat-removal line three times: stable states at 31 °C and 134 °C, an unstable one at 80 °C. The phase portrait shows which start-ups run away.

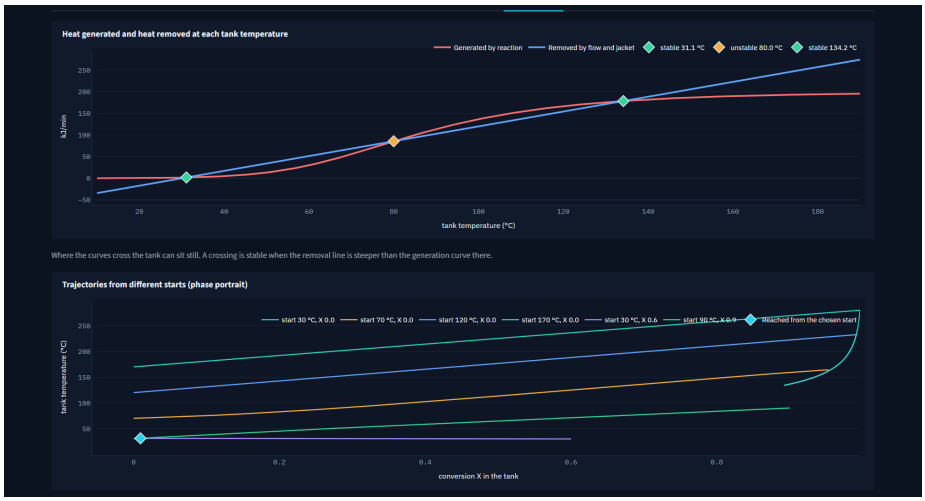


Figure 6: Energy balance: three steady states and the phase portrait ($UA = 40\,\mathrm{J/min/K}$, $E_a = 90\,\mathrm{kJ/mol}$, $\Delta H = -100\,\mathrm{kJ/mol}$).

2.3 PFR

The helical coil is treated as plug flow by default. Axial dispersion, laminar flow, recycle, adiabatic temperature rise and reaction that continues in the sample beaker are switches. Coil Reynolds and Dean numbers come from the coil geometry.

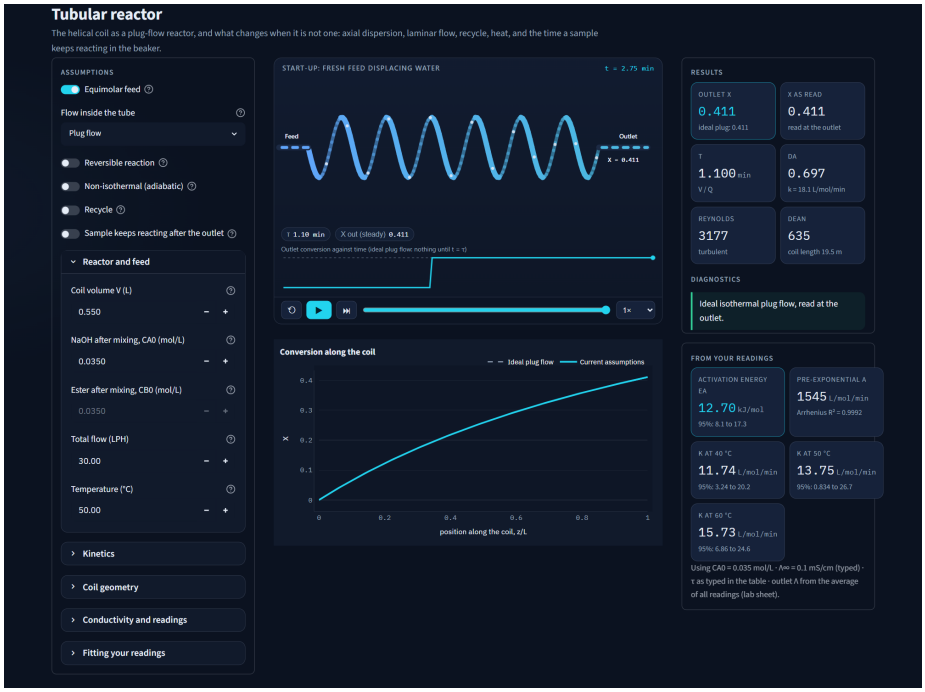


Figure 7: PFR page with fitted kinetics ($E_a = 12.70\,\mathrm{kJ/mol}$, as in the lab report).



Figure 8: PFR results: design plots, Arrhenius, conversion against flow and temperature, parity and residuals.

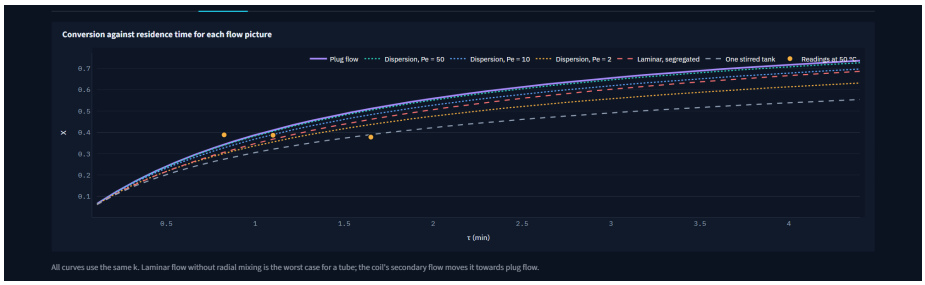


Figure 9: The same k under each flow picture, with the measured points at 50 °C.

2.4 Batch

Conductivity against time is fitted with a choice of rate law (first, second, reversible, or a fitted power). The laws are ranked by AIC. A clock offset covers the seconds between mixing and starting the stopwatch.

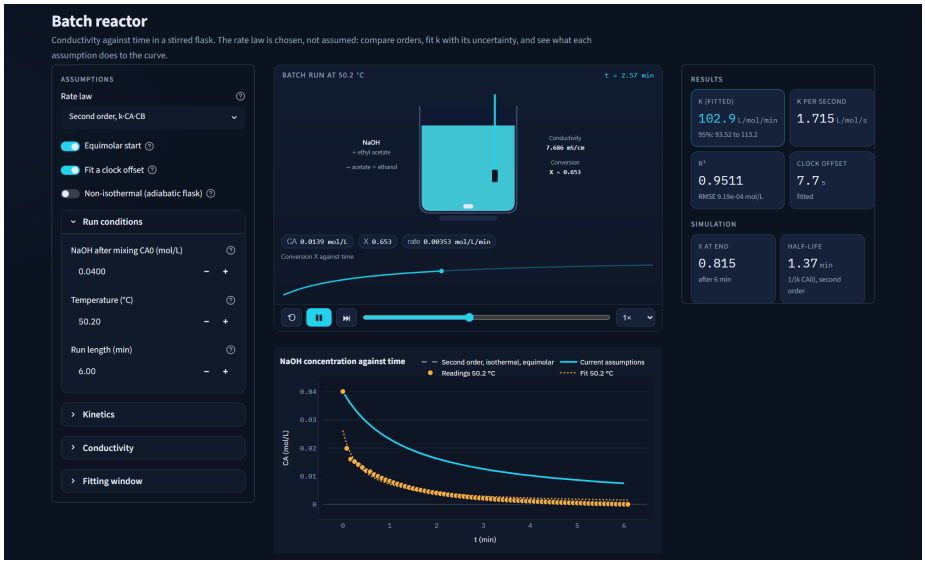


Figure 10: Batch page at 50.2 °C: fitted $k = 102.9\,\mathrm{L/mol/min}$ with its interval.



Figure 11: Integral and differential plots, residuals, and rate laws ranked by AIC ($n \approx 1.0$ by the differential method).

2.5 RTD

Up to four tracer runs share one time column. For each run the page gives t_m, σ^2, N (exact and nearest whole number), Pe under three conventions, and fitted tanks-in-series and dispersion models. It warns when the curve was cut off before the tracer had left.

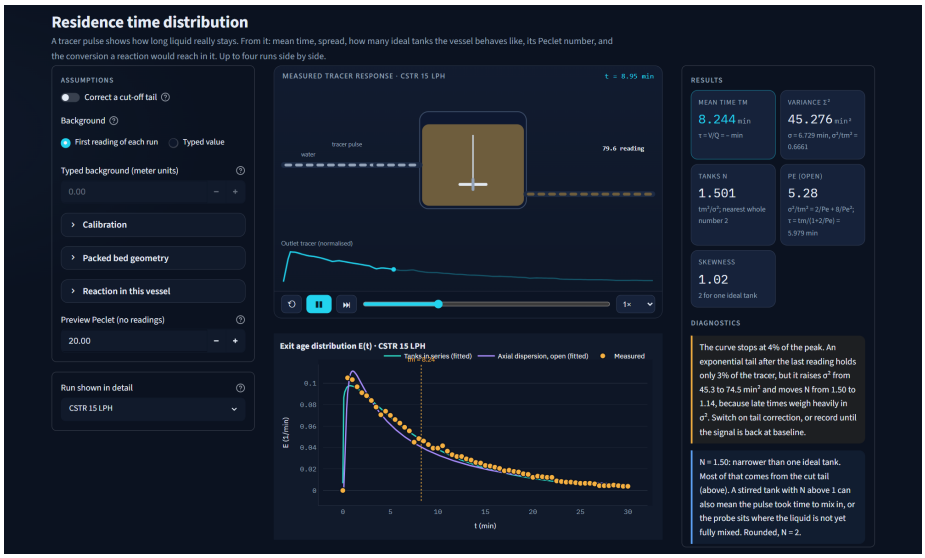


Figure 12: RTD page, stirred tank at 15 LPH. The diagnostic shows that $N = 1.5$ comes mostly from the cut-off tail.



Figure 13: All four runs: tracer curves, $E(t)$, $E(\theta)$ and $F(t)$.



Figure 14: Model fits ranked by error, with residuals of the best model.

2.6 Design Studio

The Design Studio is the "what the lab cannot do" part of the proposal: build a plant from blocks and solve it as one network, using the same rate law that was fitted on a lab page.

Blocks available on the board are Feed, CSTR, PFR, packed bed, Splitter and Product. They are dragged onto the canvas, connected by pipes, and solved together. Several feeds can enter the same tank; a splitter can send a fraction of its outlet back as recycle or around a unit as bypass. Presets load common layouts in one click: CSTR then PFR, PFR with recycle, bypassed CSTR, two feeds into a tank and a bed, and a jacketed tank with a strong exotherm.

Kinetics come from any lab page that has published a fit, or can be typed. Recycle loops are iterated with Wegstein acceleration until compositions change by less than 10^{-10} mol/L. The mass balance on sodium and acetyl groups is reported and closes to better than 10^{-8}. Each block shows its own conversion, residence time and temperature on the board; the right panel shows overall conversion, product flow, total reactor volume and the number of passes needed to converge (one or two means there is no recycle).

Below the board, six tabs open the plant up:

- **Streams** — every pipe and every unit as a table (flows, compositions, temperatures, jacket duty).
- **Profiles** — NaOH and temperature along the main path against cumulative reactor volume. A stirred tank drops straight to its outlet; a tube or bed falls gradually.
- **Levenspiel** — $1/(-r_A)$ against X along the path, so the area for each unit is its $\tau \cdot C_{A,\mathrm{in}}$.
- **Sweep** — overall conversion against one chosen parameter of one chosen block (volume, flow, splitter fraction, temperature).
- **Compare rigs** — $k(T)$ from every lab page that has published a fit, overlaid on one Arrhenius plot, plus the ideal X–Da curves for a stirred tank and for plug flow.
- **Export** — the same ZIP report as on the experiment pages.

Figure 15 shows the CSTR-then-PFR preset: feed at 0.4 L/min into a 2 L tank ($X = 0.451$ per pass), then a 0.5 L tube ($X = 0.171$ per pass). Overall conversion is 0.545, matching the sequential analytic solution, with mass-balance error below 10^{-8}.

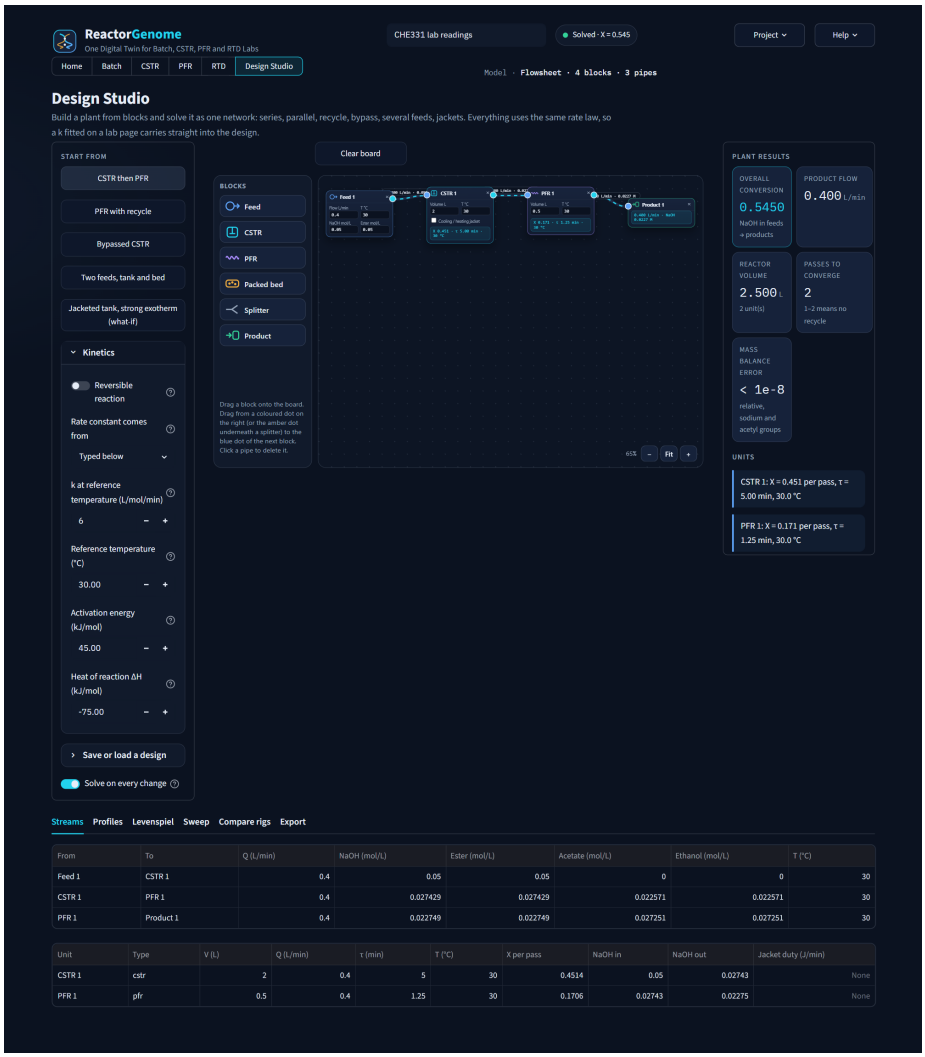


Figure 15: Design Studio: CSTR then PFR. Overall $X = 0.545$, stream table and unit table.

2.7 Export and sharing a session

Every experiment page and the Design Studio export a ZIP: an offline HTML report with the interactive graphs, CSV tables, every plot and every setting used.

The whole session also saves as one `.reactorgenome.json` project file (Project → Save). That file holds every readings table, every setting and the Design Studio board. Another student who saved their design can hand you that JSON; open it with Project → Load project. The Design Studio comes back exactly as they left it—blocks, pipes, volumes and kinetics—so groups can share plants without rebuilding them.



Figure 16: Export tab: ZIP report, HTML-only option, and the sections included.

3 Tools, model sources and tests

3.1 What was used

The app is written in Python. The pages are a Streamlit web interface, so a group opens a browser rather than installing a desktop program. NumPy and SciPy do the fitting and the reactor solves (design-plot regression, Arrhenius, ODEs along a tube, RTD moments, recycle iteration). Pandas holds the readings tables. Plotly draws the graphs. The Design Studio board is a small drag-and-drop canvas. The public copy runs on Streamlit Community Cloud at `https://che331.streamlit.app`.

The code is split in two. An engine holds only the balances and solvers. The pages call that engine and never put the equations in the buttons. Every experiment page uses the same layout: assumptions on the left, the live reactor in the middle, results on the right, the readings table underneath, then the analysis tabs.

3.2 Where the models come from

The balances are the standard ones in undergraduate reactor design, not a new rate law.

- H. S. Fogler, *Elements of Chemical Reaction Engineering*: mole balances for batch, CSTR and PFR; the design equations used for the y–τ plots; Arrhenius; non-ideal CSTR (bypass, dead volume, tanks in series); axial dispersion and the Péclet number; the energy balance and multiple steady states; RTD moments $E(t)$, $F(t)$, t_m and σ^2.
- O. Levenspiel, *Chemical Reaction Engineering*: the Levenspiel plot ($1/(-r_A)$ against X), conversion against Damköhler number, and tanks in series approaching plug flow.
- Limiting ionic conductivities (Na^+, OH^-, acetate): the default $\Lambda_\infty \approx 0.366\,\Lambda_0$, because acetate replaces hydroxide as the reaction proceeds. A typed Λ_∞ is still available.

Conductivity is turned into conversion by $X = (\Lambda_0 - \Lambda)/(\Lambda_0 - \Lambda_\infty)$. For an equimolar second-order reaction the CSTR design coordinate is $y = X/(1-X)^2$ and the PFR coordinate is $y = X/(1-X)$; in both cases $y = k\,C_{A0}\,\tau$ through the origin, so the slope is k. $k(T)$ then follows $\ln k = \ln A - E_a/(RT)$.

3.3 Tests

Each model was checked against a closed-form limit before it was shown on a page: ideal CSTR and PFR conversions, tanks in series moving toward plug flow, RTD moments of known curves, and a CSTR-then-PFR flowsheet against the same two reactors solved one after the other (mass balance below 10^{-8}). Invalid inputs and empty tables are included so a blank lab sheet does not crash a page. The suite is on GitHub: ReactorGenome / tests.